

1. Project Name & Description

- **Project Name:** SkillHub - Online Courses Management Platform
- **Project Description:**
 - **What problem does it solve?** It solves the complexity of managing online learning, enabling users to browse, filter, and enroll in technical courses, while giving administrators total control over course catalogs and user management.
 - **Target Users:** Students/Learners looking for online courses, and Administrators/Instructors managing the platform.
 - **Main Purpose:** To provide a seamless RESTful web application for online education with full authentication, authorization, and profile customization.

2. Identify Users and Roles

Role Name	Permissions	Available Actions
Admin	Full system control	Add new courses, Edit/Delete courses, Change user roles (Make Admin/User), Delete user accounts, View enrolled courses statistics.
User (Student)	Access to student features	Register/Login, Browse and filter courses by title, category, and price, Enroll in courses, Edit profile details, Upload profile picture.

3. Define Main Features

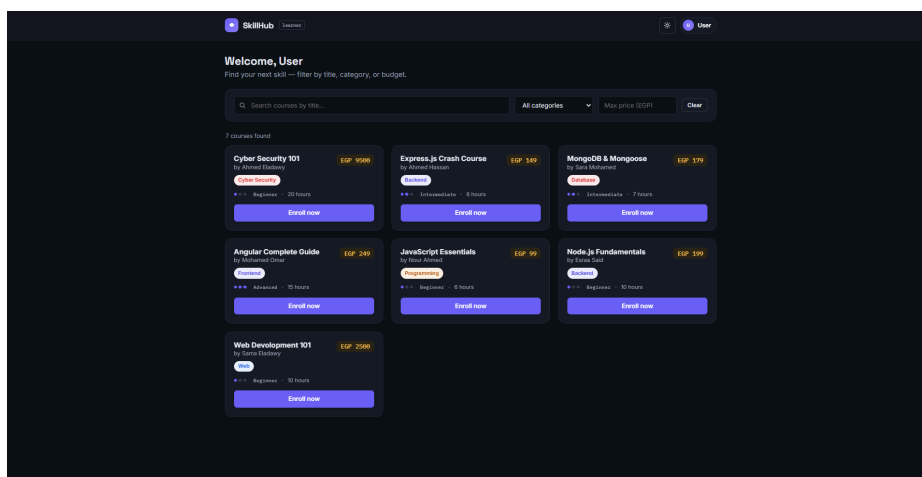
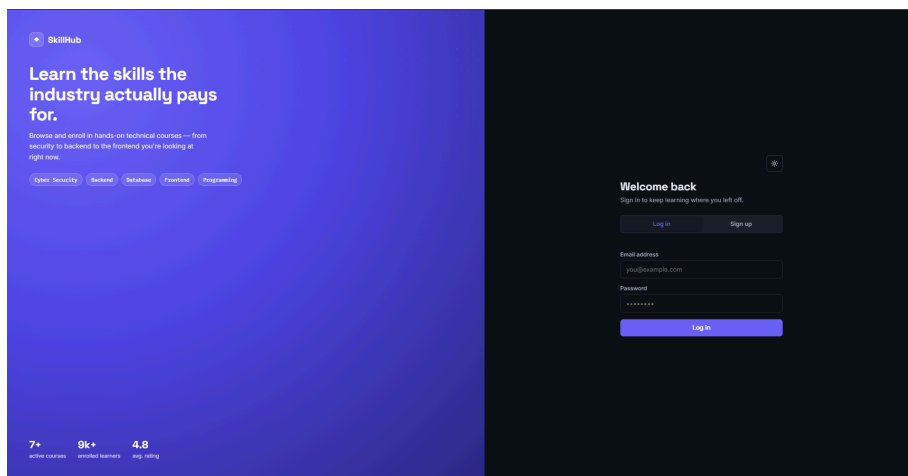
- **Authentication Features:**
 - User Registration (/signup)
 - User Login (/login)
 - Local Storage session management
- **Authorization Features:**
 - Role-based access control (Admin dashboard protection)
 - Protected routes based on user role (admin vs user)
- **CRUD Features (Courses & Users):**
 - **Courses Management:** Create (POST), Read (GET), Update (PUT), and Delete (DELETE) courses.
 - **Users Management:** Read users list, Update roles, and Delete users.
- **Image/File Upload Features:**
 - **User Profile Picture:**
 - Allowed format: Images (image/*)
 - Uploaded by: Users through device file selection (Base64 conversion).

4. Application Architecture & Tech Stack

Backend: Node.js & Express.js framework.

- **Architecture Pattern:** MVC Pattern (Models, Views/Frontend, Controllers, and Routes separation).
- **Data Storage:** JSON files (`courses-data.json` and `users-data.json`) handled via Node.js `fs` module.
- **Middleware:** `express.json()` for parsing incoming JSON bodies, custom static file serving.
- **REST API Endpoints Structure:**
 - `GET /api/v1/courses` - Get all courses
 - `POST /api/v1/courses` - Create a course (Admin)
 - `PUT /api/v1/courses/:id` - Update a course (Admin)
 - `DELETE /api/v1/courses/:id` - Delete a course (Admin)
 - `POST /api/v1/auth/signup` - Register user
 - `POST /api/v1/auth/login` - Login user
 - `GET /api/v1/auth/users` - Get all users (Admin)
 - `PATCH /api/v1/auth/users/:id/role` - Toggle user role (Admin)
 - `DELETE /api/v1/auth/users/:id` - Delete user account (Admin)
 - `POST /api/v1/auth/users/:id/enroll` - Enroll in a course (User)
 - `PATCH /api/v1/auth/users/:id/profile` - Update profile & avatar (User)

5. UI Design & Screens



SkillHub

Learn

Back to dashboard

U

Change photo

JPG or PNG — shown to admins across the platform

Full name

User

Email address

user@platform.com

Email can't be changed here.

Phone number

e.g. 070 1234 5678

Save changes

SkillHub

Admin

Log out

Admin dashboard

Manage the course catalog and your learner accounts.

COURSES

6

LEARNERS

5

ENROLLMENTS

1

ADMINS

2

Courses

Users & roles

Add a new course

Course title

e.g. React From Scratch

Instructor

e.g. Sara Mohamed

Category

Select category

Price (EGP)

e.g. 100

Level

Select level

Save course

Manage courses6 total

TITLE	CATEGORY	LEVEL	PRICE		
Cyber Security 101 Ahmed Eladawy	Cyber Security	Beginner	EGP 9500		
Express Js Crash Course Ahmed Hassan	Backend	Intermediate	EGP 5000		
MongoDB & Mongoose Sara Mohamed	Databases	Intermediate	EGP 2000		
Angular Complete Guide Omar Eladawy	Frontend	Advanced	EGP 8000		
JavaScript Essentials Nour Ahmed	Programming	Beginner	EGP 2000		
Node Js Fundamentals Eslam Said	Backend	Beginner	EGP 2500		

SkillHub

Admin

Log out

Admin dashboard

Manage the course catalog and your learner accounts.

COURSES

6

LEARNERS

5

ENROLLMENTS

1

ADMINS

2

Courses

Users & roles

Users & roles

5 total

NAME	EMAIL	PHONE	ENROLLED	ROLE	
Admin Adawy	admin@platform.com	—	0 (No courses yet)	admin	Make user
Ahmed Eladawy	adawy@gmail.com	—	0 (No courses yet)	admin	Make user
Sama Eladawy	sama@yahoo.com	—	1 (MongoDB & Mongoose)	user	Make admin
Mohamed Omar	mohamed@gmail.com	—	0 (No courses yet)	user	Make admin
User	user@platform.com	—	0 (No courses yet)	user	Make admin

